SOURCE SELECTION STATEMENT
FOR
EUROPA CLIPPER LAUNCH SERVICE

On June 8, 2021, the Source Evaluation Board (SEB) appointed to evaluate proposals for the Europa Clipper Launch Service presented the results of its evaluation to me, as the designated Source Selection Authority (SSA). Relevant portions of the SEB's evaluation and the rationale for my decision to award the Europa Clipper Launch Service based on initial proposals is set forth in this Source Selection Statement.

PROCUREMENT HISTORY

The Europa Clipper mission will conduct a detailed survey of Jupiter's moon Europa to determine whether the icy moon could harbor conditions suitable for life. The Europa Clipper will explore Europa and investigate its habitability utilizing a set of five remote sensing instruments. Key mission objectives will be to produce high-resolution images of Europa's surface, determine its composition, look for signs of recent or ongoing activity, measure the thickness of the icy shell, search for subsurface lakes, and determine the depth and salinity of Europa's ocean.

Europa Clipper requires the launch energy capability of a "Heavy-class" vehicle due to its mass and the required high-energy interplanetary trajectory. Europa Clipper has a limited 21-day launch period in calendar year 2024. The Europa Clipper launch service includes the launch vehicle, ground support facilities and equipment, test hardware and software, and all launch services required to integrate the payload to the launch vehicle and deliver the Spacecraft (SC) to its required destination.

The fiscal year 2021 Consolidated Appropriations Act, signed into law December 27, 2020, levied a requirement for NASA to utilize the Space Launch System (SLS) for the Europa Clipper if it was available and if torsional loading analysis confirmed Europa Clipper's appropriateness for SLS. If these conditions could not be met, the Administrator was to conduct a full and open competition, not limited to the launch vehicles listed in the NASA Launch Services (NLS) II contract. NASA determined both the above mentioned conditions for SLS use were not met and a full and open competition was appropriate for Europa Clipper. In response to these requirements, the Europa Clipper launch service was procured as a commercial service on an unrestricted basis (i.e., full and open competition).

Based on the requirements levied on the Launch Services Program (LSP) for the Europa Clipper launch, the NASA Flight Planning Board requested, on January 19, 2021, that LSP acquire a launch service for the mission. Europa Clipper is classified as a Class A payload, pursuant to NASA Procedural Requirement (NPR) 8705.4, Risk Classification for NASA Payloads and requires a Category 3 certified launch vehicle as currently prescribed by the NASA Flight Planning Board in accordance with NASA Policy Directive (NPD) 8610.7D, Launch Services Risk Mitigation Policy for NASA-Owned and/or NASA-Sponsored Payloads/Missions.

A sources sought synopsis was posted to the Governmentwide point of entry, beta.sam.gov, on January 26, 2021, seeking capability statements from all interested parties for the Europa Clipper launch service. Responses were received from three parties by the due date of February 8, 2021.

On February 2, 2021, a synopsis for the Europa Clipper Launch Service was posted to beta.sam.gov. The synopsis notified industry that a draft Request for Proposal (RFP) was not anticipated to be released. The synopsis requested that potential offerors notify the Government by February 17, 2021 of their intent to submit a proposal. Two potential offerors notified the Government of their intent to submit proposals by the due date. Interested parties were given an opportunity to request the draft Launch Service Interface Requirements Document (LSIRD); and to request a voluntary teleconference with the Government to review Europa Clipper requirements and to submit questions. Two parties requested the draft LSIRD and teleconferences, which were held February 9, 2021 and February 10, 2021.

Following review and consideration of industry comments to the draft LSIRD and one-on-one teleconferences, RFP No. 80KSC021R0018 was posted to beta.sam.gov on March 5, 2021, with a proposal due date of April 16, 2021. The RFP stated the acquisition was restricted to domestic launch service providers and that potential offerors should be AS9100 certified. During the course of the procurement, the contracting officer issued two amendments to incorporate changes and to respond to questions.

On March 11, 2021, the SSA appointed an SEB to evaluate proposals received in response to the solicitation.

In response to RFP 80KSC021R0018, two qualified current domestic launch service providers, Space Exploration Technologies Corporation (SpaceX) and United Launch Services, LLC (ULS), submitted timely proposals for launch services using their Falcon Heavy and Vulcan Centaur launch vehicles, respectively.

EVALUATION PROCEDURES

This acquisition was conducted as a commercial, competitively negotiated acquisition in accordance with Federal Acquisition Regulation (FAR) 12.203 in conjunction with FAR Part 15.3 and NASA FAR Supplement (NFS) Part 1815.3. This acquisition allowed for a trade-off process, as described in FAR 15.101, to obtain the best value for the Government. The RFP advised offerors it was the Government's intent to evaluate and select for award, based on initial proposals, without discussions; however, the Government reserved the right to conduct discussions if deemed in its best interest.

The RFP defined three evaluation factors, Price, Mission Suitability, and Past Performance, and provided the relative importance of each factor. The Mission Suitability factor was comprised of the following subfactors: Technical Approach, Management Approach, and Small Business Utilization.

The relative importance of the above factors for the Europa Clipper Launch Service was as follows: Mission Suitability and Past Performance factors, when combined, are significantly

more important than the Price factor. The Mission Suitability factor is more important than the Past Performance factor, which is more important than the Price factor.

The following summary from the RFP describes how Mission Suitability, Past Performance, and Price were evaluated:

Mission Suitability

The Mission Suitability evaluation was performed in accordance with FAR 15.3 and NFS 1815.3. The RFP advised that the Government would evaluate the Offeror's demonstrated understanding of the Mission Suitability subfactor requirements and approach for accomplishing those requirements, the appropriateness of the Offeror's proposed resources, and associated programmatic risk. Offerors were informed any inconsistencies identified may indicate a lack of understanding and adversely impact the Offeror's adjectival rating(s) and score. Offeror's were also informed that information and findings generated from any LSP launch vehicle certification process, NASA Space Act Agreement or other NASA contracts may be used in the evaluation of the Offeror's proposal.

The Government evaluated the consistency between all proposal volumes. The Offeror's proposed mission unique services to meet Europa Clipper's requirements were also considered in evaluating the Mission Suitability factor to further assess the risk and feasibility of the Offeror's proposed approach.

Evaluation of the Mission Suitability factor and associated subfactors focused on each offeror's technical ability, management approach, and proposed small business utilization. The Mission Suitability factor consisted of the following three subfactors:

Mission Suitability Subfactors	**Weight (Points)**
Technical Approach	700
Management Approach	250
Small Business Utilization	50
Total	**1,000**

The RFP stated that based on the above weightings, each Mission Suitability subfactor would be evaluated and then rated and scored in accordance with NFS 1815.305(a)(3)(A). This procedure required the SEB to evaluate proposals under each subfactor, to identify findings of significant strengths, strengths, weaknesses, significant weaknesses, or deficiencies; and based on these findings, assign an adjectival rating, determine a percentile score, and calculate a total point score for each Mission Suitability proposal. Each subfactor was evaluated in its entirety and the areas included in the subfactors were not individually rated, scored, or listed in a relative order of importance.

i. Technical Approach

Technical Approach included viability (feasibility and risk) of the Offeror's approach to meeting or exceeding the technical requirements listed in the Performance Work Statement (PWS) and LSIRD; an assessment of the maturity of the proposed launch

vehicle configuration and the development risk for any vehicle upgrades required to meet mission requirements; the Offeror's ability to meet all launch service performance requirements as described in Exhibit 1, Capabilities, Specifications, and Environments; an assessment of the technical justification including appropriateness, maturity, methodology and margins; and review that all capabilities and conditions stated in Exhibit 1 were consistent and compatible with all others in the proposal.

Technical Approach included the viability of the Offeror's supporting information that substantiates a flight of the booster stage, upper stage, booster engine, upper stage engine, avionics architecture, and fairing described in Table D1-C of Exhibit 1 would occur no later than 12 months before the Europa Clipper launch. The Offeror's Certification Plan was evaluated to assess the feasibility and risks associated with the Offeror's ability to satisfy the Category 3 Certification requirements of Section C, Clause 3.0, Launch Vehicle Certification; detailed schedule and data deliverables; current development status of the proposed common launch vehicle configuration; the viability of the remaining schedule for the proposed vehicle to achieve the required number of successful flights, based on the certification alternative, before L-12 months; and whether the vehicle development schedule supported the data deliverables as described in Section C, Clause 3.0, Launch Vehicle Certification.

ii. <u>Management Approach</u>

Management Approach included the Offeror's approach for program management to ensure launch success; mission schedule; approach to managing its subcontractors; plan for ensuring the safety and health of personnel and facilities; quality management plan and software assurance approach; configuration management approach for ensuring hardware and software design implementation, change process control, time/age-sensitive component control, and as built drawing control; reliability plan; risk mitigation program; manifest policy, with regard to national priority for Europa Clipper; approach to keep NASA and the payload customer informed on launch campaign progress, activities and anomalies from award through mission success determination; approach to ensuring Government insight throughout a mission flow, as well as the Offeror's approach to Government cognizance of mission unique issues.

iii. <u>Small Business Utilization</u>

Small Business Utilization included the Offeror's small business subcontracting plan and commitment to small business. The commitment to small business includes any work identified as "high technology" performed by a small business; extent of commitment to use the subcontractor (enforceable vs. non-enforceable); and the Offeror's established or planned procedures and organizational structure for small business outreach, assistance, participation in the Mentor Protégé program, counseling, market research and small business identification, and relevant purchasing procedures.

<u>Past Performance</u>

Past performance evaluations were conducted in accordance with FAR 15.305(a)(2) and NFS 1815.305(a)(2). The RFP required offerors to submit up to five past performance efforts. The RFP advised Offerors that the Government would evaluate its (and any major

subcontractors) relevant performance of work similar in size, content, and complexity to the requirements of the acquisition, and that the Government reserved the right to consider past performance information other than those identified by the offeror including, but not limited to, publicly available information related to launches, Government insight, and Contractor Performance Assessment Reports (CPARs). The RFP provided that emphasis would be given to the extent of recent direct experience and quality of past performance on previous contracts that are highly relevant to the acquisition. In evaluating past performance, the RFP advised that the Government would consider the Offeror's previously demonstrated: contract past performance, history and experience; ability to accomplish requirements to receive milestone or performance based payments; ability to perform as a launch service provider with responsibilities, including design, fabrication, vehicle and payload integration; ability to meet technical requirements and performance standards for previous work; ability to meet launch dates; approach in determining probable root cause for less than fully successful missions and resultant actions taken to improve reliability of launch services; launch vehicle history; and safety and mission assurance performance record, including any safety mishaps and associated resolution.

The RFP defined recent work as work performed in the three years prior to the proposal due date.
The SEB's overall confidence that the Offeror will successfully perform the required effort was determined by evaluating the pertinence and performance of their relevant Government and commercial contracts during the recency period. In accordance with NFS 1815.305(a)(2), offerors were assigned one of the following confidence ratings: Very High Level of Confidence, High Level of Confidence, Moderate Level of Confidence, Low Level of Confidence, Very Low Level of Confidence, or Neutral.

Price
For the Price Factor, analysis was performed in accordance with FAR 15.404-1(b) to determine the reasonableness of the proposed pricing. Certified cost pricing data was not requested nor required.

In accordance with the RFP, the total evaluated price (TEP) was also considered. The TEP included the following:

CLIN 1 Europa Clipper Launch Service
SubCLIN 1A Standard Launch Service and Standard Mission Integration
SubCLIN 1B Mission Unique Services

SEB EVALUATION

The evaluation board conducted all evaluations using the above summarized evaluation criteria as specified in the RFP. In conducting its evaluations, the evaluation team utilized dedicated evaluators from appropriate disciplines to provide specific expertise needed in the evaluation process. Using the findings and analyses of the evaluators, the team generated evaluation consensus findings, which identified and assessed significant strengths, strengths, weaknesses,

significant weaknesses, and deficiencies for each proposal accordingly. In addition to the evaluation of the factors and subfactors identified above, the evaluation team ensured all solicitation requirements established by the RFP were met. Furthermore, the SEB reviewed for completeness each offeror's administrative data, which consisted of a completed Standard Form 1449; Representations, Certification, and Other Statements of Bidders; Model Contract; and Statement of Acceptance/Summary of Exceptions.

The SEB utilized the following definitions for identification of strengths, weaknesses, and deficiencies:

> "Deficiency" is a material failure of a proposal to meet a Government requirement or a combination of significant weaknesses in a proposal that increases the risk of unsuccessful contract performance to an unacceptable level.
>
> A "Significant Weakness" in the proposal is a flaw that appreciably increases the risk of unsuccessful contract performance.
>
> "Weakness" means a flaw in the proposal that increases the risk of unsuccessful contract performance.
>
> "Strength" is an aspect of the proposal that will have some positive impact on the successful performance of the contract.
>
> "Significant Strength" is an aspect of the proposal that greatly enhances the potential for successful contract performance.

Mission Suitability:

The substance of the SEB's evaluation of each offeror's proposal under the Mission Suitability factor is summarized below (in alphabetical order):

SpaceX

SpaceX's proposal received the highest overall Mission Suitability score. The SEB identified one strength and nine weaknesses in the proposal. There were no significant strengths or significant weaknesses identified. The following is a summary of the SEB's evaluation of SpaceX's proposal under the three Mission Suitability subfactors:

Mission Suitability – Technical Approach: The SEB rated SpaceX's proposal as Good under this subfactor. The SEB identified the following strength and weaknesses (presented at a summary level):

STRENGTH:

- Flight-Correlated Analyses and Baseline Launch Vehicle Capability

WEAKNESSES:

- Avionics Configuration
- Telemetry Format Changes
- Figure of Merit Calculation
- Molecular Deposition from Launch Vehicle Sources
- Spacecraft Flight Plug Installation Timeline and Contingency Operations
- Minimizing Unplanned Environmental Control System Outages
- Mission Unique Services Basis of Estimates

There were no significant strengths, significant weaknesses, or deficiencies assigned.

Mission Suitability – Management Approach: The SEB rated SpaceX's proposal as Good under this subfactor. The SEB identified the following weaknesses (presented at a summary level):

WEAKNESSES:

- Booster Fleet Management
- Work Plan Hardware Assignments and Certification Events

There were no significant strengths, strengths, significant weaknesses, or deficiencies assigned.

Mission Suitability – Small Business Utilization Subfactor: The SEB rated SpaceX's proposal as Good under this subfactor. The SEB did not identify any significant strengths, strengths, deficiencies, significant weaknesses, or weaknesses.

ULS

ULS's proposal received the lowest overall Mission Suitability score. The SEB identified one deficiency, four significant weaknesses, and twelve weaknesses in the proposal. There were no significant strengths or strengths identified. The following is a summary of the SEB's evaluation of ULS's proposal under the three Mission Suitability subfactors:

Mission Suitability – Technical Approach: The SEB rated ULS's proposal as Poor under this subfactor. The SEB identified the following deficiency and significant weaknesses described below, and weaknesses presented at a summary level:

DEFICIENCY:

- Ability to Achieve First Flight of Vehicle Block Upgrades Prior to L-12 Months – A requirement for the Europa Clipper launch service is all block upgrades, whether for the fleet or Europa Clipper specific, that are not considered common launch vehicle configuration changes shall have at least one flight no later than 12 months (L-12) before the Europa Clipper launch. This includes the booster stage, upper stage, booster engine, upper stage engine, avionics architecture, and fairing. In order to meet the Europa Clipper mission requirements, [redacted] This is a material failure of the proposal to meet a critical risk reduction requirement for a Class A, NASA flagship mission.

  In addition, the proposal does not identify [redacted] as Mission Unique Services (MUS). However, these meet the definition of a MUS and therefore, the proposed approach prevents Government approval of the design through the mission specific design review process.

  Finally, the proposal did not define the BE-4 engine block version proposed. Booster engine design or operational changes that drive requalification create an increased risk of schedule delays.

SIGNIFICANT WEAKNESSES:

- Performance Capability – ULS proposed that through the [redacted] of Vulcan Centaur it would be able to meet the Europa Clipper mass requirements; however, the proposal does not provide substantiating information that describes the performance benefit for [redacted]. Europa Clipper has a mass of 6,065kg and the current version of the Vulcan Centaur A/B has approximately 5,000kg of performance capability to the Europa Clipper targets. Additionally, the proposal does not describe the performance reserve methodology. Lack of detail on performance reserve methodology and vehicle upgrades with undefined performance benefits leads to uncertainty in the Vulcan Centaur mass-to-orbit capability. Uncertainty in launch vehicle performance capability significantly

increases the risk that the launch vehicle will not be able to lift the Europa Clipper mass to the required orbital target which would result in mission failure.

- Dynamic Uncertainty Factor on Limit Loads – ULS proposed a dynamic uncertainty factor (DUF) of [redacted] for the Vulcan Centaur spacecraft center of gravity (CG) limit load factors, which has not achieved first flight yet. Applying a DUF of only [redacted] for an unflown launch vehicle increases the risk the Europa Clipper spacecraft CG limit load factors will be exceeded and ULS did not provide sufficient rationale for why a DUF of [redacted] was appropriate. Since the Europa Clipper spacecraft is being manufactured, there are no opportunities for spacecraft design changes to correct for a change in CG limit load factors should they be required after Vulcan Centaur's first flight. Therefore, there is an appreciable risk to achieving the 2024 planetary launch period.

- Initial Integrated Thermal Analysis Timeline – The ULS proposal included the initial Thermal Analysis deliverable at receipt of payload model +6 months instead of the required receipt of payload model +4 months. The Europa Clipper flight hardware is currently being manufactured and confirmation of the compatibility of the thermal environment prior to environmental testing is crucial because of the maturity of the spacecraft hardware and software. The spacecraft requires the results of the initial thermal analysis as soon as possible, as any late identification of non-compliance to the spacecraft's thermal limits increases the likelihood of the spacecraft having to take additional thermal protective measures which could result in both hardware and software impacts. Receiving the initial thermal analysis two months later than required appreciably increases the risk to the spacecraft integration and test timeline activities.

- Feasibility of Certification Schedule – ULS proposes certification of the Vulcan Centaur through the Category 3, Alternative 2 method. This requires six flights of the common launch vehicle configuration. ULS proposes to achieve the sixth flight [redacted] prior to Europa Clipper launch, leaving [redacted] to complete certification by the L-12 month requirement.

  The provided certification schedule is not viable for the vehicle to achieve six flight before L-12 months. No consideration for launch delays driven by launch vehicle development, initial and subsequent flight delays, flight anomalies, failures, and spacecraft delays were included.

  Completion of certification after L-12 months increases the likelihood that significant technical risks cannot be mitigated and must either be accepted or cause mission delays that threaten the Europa Clipper planetary launch period.

WEAKNESSES:

- Injection Covariance Matrix
- Acoustic Environment
- Shock Environment
- Random Vibration Environment

- Environmental Control System Redundancy during Transportation and Handling
- Payload Compartment Silicone Usage and Material Data Sheets
- Down Range Telemetry Services
- Mechanical Interface Compliance
- Trade Secret Information Barriers to Insight
- Work Plan Certification Events
- Europa Clipper Radiator Thermal Conditioning

There were no significant strengths or strengths assigned.

Mission Suitability – Management Approach: The SEB rated ULS's proposal as Good under this subfactor. The SEB identified the following weakness (presented at a summary level):

WEAKNESS:

- Detailed Mission Schedule

There were no significant strengths, strengths, significant weaknesses, or deficiencies assigned.

Mission Suitability – Small Business Utilization: The SEB rated ULS's proposal as Good under this subfactor. The SEB did not identify any significant strengths, strengths, deficiencies, significant weaknesses, or weaknesses.

Past Performance:

The substance of the SEB's evaluation of each offeror's proposal under the Past Performance factor is summarized below (in alphabetical order):

SpaceX

Based on SpaceX's performance on recent and relevant contracts, the SEB has a High Level of Confidence SpaceX will successfully perform Europa Clipper's mission requirements based on SpaceX's performance on recent and relevant contracts.

SpaceX's past performance is very highly pertinent to this acquisition, in terms of size, content and complexity with relevant experience in launch service contract terms, conditions, and scope and providing launch services. SpaceX's past performance citations demonstrate, overall, very effective performance in the areas of quality, schedule, and management. Past performance for work performed under the NLS II, Evolved Expendable Launch Vehicle, and National Security Space Launch contracts, for critical government science and defense payloads were considered the most relevant. For these contracts, SpaceX successfully executed requirements that were the same or very similar in terms, conditions, and scope, utilizing the same or very similar launch vehicles. In addition, SpaceX successfully performed relevant and somewhat relevant services under the Commercial Crew Transportation Capability, Commercial Resupply Services, and commercial launch contracts, further contributing to the overall high level of confidence rating.

Under contracts assessed, SpaceX consistently met requirements to receive milestone or performance-based payments. Additionally, SpaceX regularly submitted Contract Data Requirements List Deliverables (CDRLs) and other technical and business reports either on time or early. SpaceX also demonstrated an ability to meet milestones or recover schedule when customer provided input was delayed. While there were instances where the delivery of CDRLs were late or inadequate, these instances were noted early in the evaluation period and were not frequently occurring events. Therefore, these instances did not detract from the overall performance rating.

SpaceX demonstrated excellent performance as a launch service provider across several government contracts all involving design, fabrication, vehicle, and payload integration. Early in the evaluation period, a [redacted]. The Government is confident in SpaceX's ability to meet design, fabrication, vehicle, and payload integration requirements. In addition, while SpaceX has demonstrated the ability to meet strict environmental and contamination control requirements on GPS missions, the SEB found that it has not had an interplanetary mission that imposes additional integration challenges associated with planetary protection requirements.

SpaceX demonstrated excellent performance in meeting technical requirements and performance standards for previous work. As documented in several CPARS, [redacted] However, a vast majority of SpaceX's CPARS ratings under quality were "Very Good" or "Exceptional," which demonstrates SpaceX's commitment to providing technical solutions that meet its customers' requirements.

While SpaceX has experienced some challenges to meeting launch dates, LSP's past experience has provided insight regarding this concern. SpaceX has demonstrated its willingness and ability to surge resources to prioritize government launches including rescheduling lower priority missions in favor of higher priority government missions. Additionally, Falcon Heavy is currently in operational service with no proposed block upgrades required to meet Europa Clipper's mission requirements, therefore the SEB has confidence in SpaceX's ability to meet the Europa Clipper launch window.

SpaceX demonstrated superior performance in investigating anomalous vehicle performance, determining probable root causes, and taking actions to improve reliability of launch services. SpaceX has a 100 percent success record on all launches for the Department of Defense (DoD), NASA, and its commercial customers during the evaluation period. LSP has experience working with SpaceX through their anomaly process and has been an invited member of their anomaly investigation teams.

During the evaluation period, SpaceX successfully launched 63 out of 63 missions using the Falcon 9 FT and Falcon Heavy launch vehicles. Of the 63 total missions launched, two were of the Falcon Heavy configuration, the proposed common launch vehicle configuration for the Europa Clipper mission, and 38 were for customers (i.e. not SpaceX Starlink launches). The Falcon Heavy is still early in its demonstrated flight history with three successful flights. This is mitigated somewhat by its extensive hardware commonality with the Falcon 9 FT vehicle, which has significant flight history.

Finally, SpaceX has demonstrated an effective safety and mission assurance performance record. SpaceX employs a thorough and rapid investigation process for safety mishaps and anomalous launch vehicle performance in test and flight.

ULS

Based on ULS's performance on recent and relevant contracts, the SEB has a High Level of Confidence that ULS will successfully perform Europa Clipper's mission requirements

ULS's past performance is highly pertinent to this acquisition, in terms of size, content and complexity with relevant experience in launch service contract terms, conditions, and scope and providing launch services. ULS's past performance citations demonstrate, overall, very effective performance in the areas of quality, schedule, and management. Past performance for work performed under the NLS II, Solar Probe Plus, EELV, and NSSL contracts, for critical government science and defense payloads are considered relevant. For these missions, ULS successfully executed requirements that are the same or very similar in terms, conditions, and scope on different CLVCs. In addition, ULS has also successfully performed somewhat relevant services under the Orbital Test Flight commercial launch contract, further contributing to the overall highly pertinent determination with very effective performance.

Across all evaluated contracts, ULS consistently met requirements to receive milestone or performance-based payments. Numerous government customers cited early or on-time delivery of most contract deliverables, with only minor issues noted. While there were a few instances where the delivery of CDRLs was late or inadequate, these instances were noted early in the evaluation period and did not detract from the overall performance rating.

ULS demonstrated excellent performance as a launch service provider across several government contracts all involving design, fabrication, vehicle, and payload integration. For Europa Clipper, ULS proposes its Vulcan Centaur launch system, which is currently in non-recurring development, and is targeting first flight in calendar year 2022. Since the Vulcan Centaur launch system has no flight history, the SEB used information from the Atlas and Delta missions launched during the evaluation period for past performance to assess ULS's ability to perform as a launch service provider. During the evaluation period, ULS demonstrated a stable concept of operations for vehicle and payload integration and the ability to integrate and launch large solitary payloads with masses that exceed those of the Europa Clipper spacecraft. The planned Vulcan Centaur concept of operations for payload integration is similar to the Atlas V with both occurring in the ULS Vertical Integration Facility. ULS has demonstrated the ability to meet strict environmental, contamination control, and planetary protection requirements for

interplanetary missions including Mars 2020 and InSight which were both missions to the surface of Mars with similar planetary protection requirements to Europa Clipper.

ULS demonstrated exceptional performance in meeting technical requirements and performance standards for numerous government and commercial customers. Although there were some cited lapses in quality during launch site operations for some missions, ULS's corrective actions were effective and issues were quickly resolved in most cases.

ULS has demonstrated exceptional ability to meet launch dates for the government's high-value, high-priority payloads. Overall ULS has shown its ability to meet launch dates on the 15 missions launched during the evaluation period including four LSP limited launch period missions, demonstrating ability to meet interplanetary launch period requirements. The initial Vulcan Centaur launch has experienced delays; however, this is commensurate with historical vehicles in the launch industry.

ULS has demonstrated superior performance in identifying anomalies, determining probable root causes, and taking actions to improve reliability of launch services. ULS has a 100 percent success record on all launches for the DoD, NASA, and its commercial customers during the evaluation period. LSP has experience working with ULS through their anomaly process and has been an invited member of their anomaly investigation teams. The ULS lessons learned process, including its Data Observation and Data Anomaly identification and resolution process, is well established, facilitates resolution, and reduces future risk.

The SEB considers ULS's launch vehicle history as a company to be exceptional; however the Vulcan Centaur is unproven as a launch vehicle as it is still in protracted development. Although there is some commonality with the Atlas V and Delta IV launch vehicles, the commonality of launch environments and operating conditions cannot be fully asserted until a first flight is achieved. The lack of launch history for the proposed common launch vehicle configuration is a concern; however, it did not have a significant impact on the overall level of confidence as a result of ULS's launch vehicle history as a company. During the evaluation period, ULS successfully launched 15 out of 15 missions using the Atlas and Delta family of vehicles. Of the 15 total missions launched, nine were Atlas V, two were Delta IV Medium, and three were Delta IV Heavy, all of which have similarity to the proposed Vulcan Centaur. One mission was the final Delta II, which is a different common launch vehicle configuration with very little similarity to Vulcan Centaur.

ULS has demonstrated an effective safety and mission assurance performance record. ULS employs a thorough investigation process for safety mishaps and anomalous launch vehicle performance in test and flight with consistent and open communication with government customers including invitations to Corrective Action Boards and Anomaly Review Boards. ULS's investigations and implementation of corrective and preventative actions have been demonstrated to be correct, thorough, and effective.

ULS cited five major subcontractors for various components of the Vulcan Centaur rocket (Aerojet Rocketdyne – RL10 engines; Blue Origin – BE-4 booster engines; L3 Harris – avionics; Northrop Grumman Space Systems – GEM 63XL solid rocket boosters; and RUAG Space USA

– payload fairing, heatshield, and interstage adapter). A review in the CPARS for each of the major subcontractors did not produce any relevant performance information or records consistent with the subcontractor's responsibilities delineated in the proposal. The SEB reviewed ULS's CPARS for any mention of the identified major subcontractors' components and performed an evaluation of the available information. In addition, an internet search was performed for each of the identified major subcontractors and the component being provided. No adverse information was found regarding any of the identified major subcontractors or components.

Price:

In accordance with RFP Section E3, 4.0, Price Factor, the SEB utilized each offeror's total evaluated price to conduct price analysis. SpaceX's total overall proposed price is $178,322,196 which includes the Standard Launch Service and Standard Mission Integration Service and all Mission Unique Services. ULS's overall total evaluated price is substantially higher than SpaceX's.

In accordance with RFP Section E2, 8.4, Financial Capability, the SEB assessed each offeror's proposal for financial capability. Both offerors were determined to be financially capable.

Based on adequate price competition, the total evaluated prices for SpaceX and ULS were determined to be fair and reasonable.

DECISION

On June 8, 2021, the Europa Clipper Launch Service SEB presented the results of their evaluation to me. During the presentation, I was fully briefed on the procurement process and was given detailed evaluation materials concerning the proposals submitted for the Europa Clipper launch service competition. I questioned the SEB members on the material presented and carefully considered the detailed mission suitability findings, past performance information, and pricing data presented by the SEB. Additionally, I solicited and considered the views of key senior personnel within Kennedy Space Center leadership and LSP who participated in the review. These key senior personnel thoroughly understood the application of the evaluation factors set forth in the RFP, due to their current job responsibilities as related to the Europa Clipper procurement and/or their extensive experience with other similar procurements. Soliciting additional comments or questions from these senior management officials yielded no significant issues or stated concerns. In determining the possibilities for moving forward, namely, of either awarding on initial proposals, as is contemplated in the RFP, or alternatively, establishing a competitive range consisting of one or both offerors, I referred to the following relative order of importance for the three evaluation factors specified in the RFP:

Mission Suitability and Past Performance factors, when combined, are significantly more important than the Price factor. The Mission Suitability factor is more important than Past Performance factor, which is more important than Price factor.

The selection rationale that follows was based on a comparative assessment of the proposals against each of the three source selection factors.

I began by reviewing the findings presented by the SEB within the Mission Suitability factor for each offeror and reading each of the detailed findings. While this decision is a product of my independent judgement, my review of each finding under this factor leads me to adopt them as my own. I believe the evaluation of the proposal in this area has been comprehensive, thorough, and well documented. Any differences or nuances in how I view the findings generated under these factors will be discussed below. The SEB's clear findings enabled me to determine that awarding on initial proposals was merited given the reasonably sound response demonstrated under the SpaceX Mission Suitability proposal, the significant qualitative difference in Mission Suitability proposals, and the substantial difference in price.

As a preliminary matter, when I reviewed the content across the Offerors' Mission Suitability proposals, each had varying degrees of risk assessed, with the most egregious being the deficiency and two of the significant weaknesses (i.e., Performance Capability and Feasibility of Certification Schedule) assessed under ULS's proposal. This is a significant discriminator between the two proposals. The deficiency constitutes a material failure of the proposal to meet Government requirements, and increases the risk of unsuccessful contract performance to an unacceptable level. Based on my judgement, the deficiency and the two significant weaknesses cannot be satisfactorily addressed during discussions. I considered the relative order of importance of the RFP evaluation factors and in light of the significant mission suitability discriminators and the significantly higher price and despite ULS's High Level of Confidence rating for past performance, I conclude that ULS is not competitive for award. Therefore, the discussion below is solely on the remaining proposal.

The SEB identified nine weaknesses and one strength in SpaceX's Mission Suitability framework. The resultant weaknesses captured by the SEB represent the unique challenges of this mission. Despite the weaknesses, the Europa Clipper RFP allows for award without discussions. I agree with the SEB's findings and find that the weaknesses identified in is this area of the proposal present minimal risk to acceptable execution of this contract. SpaceX proposed a common launch vehicle configuration which has three successful flights and does not require vehicle upgrades to meet the mission trajectory requirements. These are appreciable mitigations for technical and schedule risk. Therefore, I agree with the SEB's assessment for the Mission Suitability factor.

I considered the SEB's evaluation of past performance and their explanation of how they considered each element of past performance stated in the RFP. Accordingly, I agree with the SEB's level of confidence rating of High for past performance. SpaceX's past performance is very highly pertinent to this acquisition, in terms of size, content and complexity with relevant experience in launch service contract terms, conditions, and scope and providing launch services on the proposed CLVC and other launch vehicles. SpaceX's past performance citations demonstrate overall very effective performance in the areas of quality, schedule, and management. Though they have only a few launches in the vehicle configuration proposed for this mission, their demonstrated repeated successful launches across all configurations provide me with a high confidence in their technical capabilities.

The price evaluation considered whether the proposed prices are reasonable for the launch service described in the proposal. At the presentation, the SEB explained its analysis of the proposal in detail, including assessment of the proposed launch vehicle and its prices in relation to the independent government cost estimate, current market conditions, previously competed prices on the current NLS II contracts, and other vehicles in the same class. SpaceX adequately supported the proposed prices for its vehicle and mission unique services in the current market. I find the proposal offers a price that is fair and reasonable for the launch service proposed.

I understand the relative order of importance of the evaluation factors specified in the RFP and that I have the ability to trade-off in the award selection when it may be in the best interest of the Government to consider award to other than the lowest priced offeror where mission suitability and past performance are more important than price. In the present circumstances, only one proposal has been deemed acceptable and that proposal also had the lowest price.

Accordingly, after considering the above cited selection criteria, the findings of the SEB, and exercising my independent judgment, I hereby select SpaceX as the best value for the award of the contract for the Europa Clipper Launch Service.

AMANDA MITSKEVICH Digitally signed by AMANDA MITSKEVICH Date: 2021 07.15 06:41:17 -04'00'

Amanda M. Mitskevich
Manager, Launch Services Program
Source Selection Authority